

HANOI UNIVERSITY OF MINING AND GEOLOGY
INFORMATION TECHNOLOGY

ENGLISH FOR DATA SCIENCE
CLASSNOTE

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ENGLISH FOR DATA SCIENCE

How to use this book

This book is designed for students first time learning and approach Data Science English terms and techniques. Students are not required to be proficient in English, but are able to catch and memorize definitions and contexts to use Data Science words relevantly. Furthermore, students are expected to be able read and understand English referenced materials and tutorials in the discipline of Data Science. This book mainly focuses on ***Vocabulary and Reading Comprehensive Skills***. Students are encouraged to take extra English documents and practices from the Internet or other resources, even AI supportive modules or apps in advance.

This book consists two parts:

- 08 DS topics in hand: Students will be involved in 08 most common Data Science topics, such as, definition of Data Science, skills and tools for Data Science, algorithms and statistics, and DS applications, etc.
- Data Science Industrial Glossary: a capsuled wordlist of common terms that students need to get on in order to work and research in Data Science.

The structure of each lesson: each lesson (except the Glossary) will be divided into 3 sections:

- Section 1 – **Vocabulary**: there is a list of new words/ terms following up with the assigned topic. There, students need to find out the meaning of each word/term and its spell and pronunciation. ***Students need to prepare this part at home.***
- Section 2 – **Reading Comprehensive**: A related topic text will be provided. In this section, students will be able to scan main ideas, perceive definitions, skim details, and understand the context. Students are also encouraged to prepare beforehand. Texts are referenced from “Data Science for Beginners – Comprehensive guide to most important basics in Data Science” by Alex Campbell (copyright).
- Section 3 – **Developing skills**: there are several exercises for students to look over all that discussed in the topic. There are mixed types, such as: matching question, True/False questions, understanding questions or simple questions answering.

About the Glossary: the industrial terms are organized following up the Alphabet system with approximately 88 terms and their short descriptions. This material is referenced from the “Data Science Glossary” made by Pragmatic Institute, LLC (copyright).

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TOPIC 1

A BRIEF OF DATA SCIENCE AND DATA SCIENTIST

I. Vocabulary

Word	Meaning	Word	Meaning
Data Science		Sensor	
Algorithm		Natural Calamity	
Tool		Customer Service	
Principle		Browsing History	
Variable		Purchase History	
Pattern		Mathematical Model	
Data Set		Statistical Model	
Outcome		Data Scientist	
Data Analyst		Function	
Insight		Business Question	
Probability		Scientific Discipline	
Predictive Analytics		Recommendation	
Prescriptive Analytics		Forecasting	
Structured Data		Instrument	
Unstructured Data		Insights	
Semi-Structured Data		Domain	
Business Intelligence		Application	
Self-Driven Car		Calamity	

II. Reading text

Introduction to Data Science

Data science is a mix of numerous algorithms, tools, principles, and languages to identify the hidden patterns within the variables in the data set. This may lead you to wonder how this is different from what has been done on data for years. The answer is that earlier we could only use tools and algorithms to explain the variables in the data set, but using data science, it becomes easier to predict the outcomes.

A data analyst uses the data only to explain what is happening in the present using historical data set. On the other hand, a data scientist only looks at the data to obtain any insights from the

data set. He also uses some advanced algorithms to identify the probability of the occurrence of an event. He looks at the data from various angles and aspects.

Data science is used to make informed decisions based on predictions made using the existing data set. You can apply numerous analytics to the data set to obtain this information. If you want to develop a model predict the possibilities or outcomes of a futuristic event, you need to use predictive causal analytics. You may need to use a model that can take the required decisions and modify the parameters based on the data set or question. To do this, you need to use prescriptive analytics. This form of analytics is more about providing the right information so that you can make an informed decision. You can also use this type of analytics to predict a range of associated outcomes and prescribed actions.

Why Use Data Science?

Before organizations collected data from every device used, they worked with small volumes of data. It was easy to analyze and understand the data and relationships within the data set using some business intelligence tools. Most traditional business intelligence tools only worked on structured data sets, but most of the data collected today is either semi-structured or structured. It is important to understand that most data collected now are semi-structured or unstructured.

Simple business intelligence tools cannot process this type of data, especially since large volumes of data are collected from different instruments. It is for this reason we need to develop advanced and complex analytical algorithms and tools to process, analyze and draw some insights from the data. It is not only for this reason why data science has gained popularity. Let us look at how data science is used in different domains.

Customer Service

How great would it be if you could know exactly what your customers want? Do you think you can use existing data to learn more about your customers, such as purchase history, browsing history, income, and age. You may have had this data with you in the past, as well. Since you use different mathematical and statistical models, you can effectively work with large volumes of data and identify the right products to recommend to your customers. This is a great way to bring more business to your firm.

Self-Driven Cars

How would you feel if your car could drive you home? Numerous companies are trying to develop and improve the workings of a self-driven car. The cars collect live information from various sensors, such as lasers, radars, and cameras. To create a map of the surrounding environment. The algorithm in the car uses this data to decide to speed up, slow down, park, stop, overtake, etc. These algorithms are often machine learning algorithms.

Predictions

Let us now consider how you can use data science in predictive analytics. Consider weather forecasting. The algorithms used take data from aircraft, satellites, radars, ships and other parts to collect and analyze data. This helps you build the required models. You can use these models to predict the occurrence of any natural calamities. Using this information, you can take the necessary measures to save lives.

Who is a Data Scientist?

If you look for the words data scientist on the Internet, you may come across numerous definitions. A data scientist uses data science to answer some business questions and concerns. The term data scientist was coined when people learned that a data scientist uses data, various mathematical or statistical functions and operations, and other scientific fields and applications to make sense of the data in the database.

Functions Performed by Data Scientists

A data scientist is one who cracks various data problems using their expertise in specific scientific disciplines. He works with different mathematical, statistical, and computer science elements. He does not necessarily have to be an expert in these fields. That said, he would use some technologies and solutions to develop the right solutions and reach conclusions crucial for the organization's development and growth. A data scientist finds a way to present the data in a useful form when compared to the data available in the data set. They work with both structured and unstructured data. Let us now look at what business intelligence is, and how it is different from data science. You may have heard about business intelligence, and most people confuse data science with business intelligence. We will look at some differences between the two, so you understand better.

NOTES:

III. Exercises

1. Exercise 1: Vocabulary Matching

Match the terms in Column A with their correct definitions in Column B.

Column A

- i. Data Science
- ii. Predictive Analytics
- iii. Prescriptive Analytics
- iv. Data Scientist
- v. Business Intelligence

Column B

- A. Analyzing historical data to explain current patterns.
- B. A discipline involving tools and algorithms to extract insights and predict outcomes.

- C. A method for predicting possible future outcomes and associated actions.
- D. A professional who uses data to answer business questions and provide insights.
- E. Techniques to recommend actions based on data predictions and decisions.

2. Exercise 2: True or False

Read the statements below and mark them as **True** or **False**.

- a. Traditional business intelligence tools can easily handle unstructured data.
- b. A data scientist and a data analyst perform the same roles.
- c. Self-driven cars use machine learning algorithms to make decisions.
- d. Predictive causal analytics helps in predicting the outcomes of future events.
- e. Data science can help recommend products to customers based on purchase history.

3. Exercise 3: Fill in the Blanks

Complete the sentences using the correct words from the text.

- a. A _____ uses advanced algorithms to identify patterns and probabilities in data.
- b. _____ analytics focuses on making informed decisions by modifying parameters in a data set.
- c. Most traditional business intelligence tools work best with _____ data.
- d. Self-driven cars rely on data collected from sensors such as _____ and cameras.
- e. Weather forecasting is an example of _____ analytics in action.

4. Exercise 4: Short Answer Questions

Answer the following questions in 1–2 sentences.

- a. How does data science differ from traditional data analysis?
- b. What is the role of a data scientist in an organization?
- c. Why are advanced analytical tools necessary for modern data processing?
- d. Give an example of how data science is used in customer service.
- e. What type of data do self-driven cars use to make decisions?

5. Exercise 5: Discussion

Write a short paragraph (5–6 sentences) to answer the following question:

What are some benefits of using data science in predicting natural calamities?

Include examples from the text to support your answer.

TOPIC 2

DATA SCIENCE LIFE CIRCLE

I. Vocabulary

Word	Meaning	Word	Meaning
Lifecycle		SQL	
Discovery		Hadoop	
Business Requirements		Training Data	
Specifications		Testing Data	
Hypothesis		Clustering	
Data Preparation		Classification	
Analytical Sandbox		Association	
Extract, Transform, Load (ETL)		Production Environment	
Outliers		Constraints	
Exploratory Data Analysis (EDA)		Key Findings	
Modeling		Insights	
Algorithm		Real-time Data	
Variables			

II. Reading text

Data Science Lifecycle

Phase One: Discovery

Before you work on the project, you need to understand the following:

- Business requirements
- Specifications
- Required or approved budget
- Priorities

If you want to pursue a career in data science, you need to possess the ability to ask important questions. You need to assess if you have the right resources, people, technology, data and time to support the work done on the project. This is the phase where you frame the problem and identify the initial hypothesis you want to test.

Phase Two: Data Preparation

When you identify the required resources needed to work on the analysis, you need to develop or identify an analytical sandbox where you can perform the testing and analysis of the data. You need to process, explore and condition the data before you model it. You also need to perform the following operations to move the data into the sandbox environment:

- Extract
- Transform
- Load
- Transform

Most data scientists use R or Python to clean transform and visualize the data used in the analysis. These programming languages help you to identify the outliers in the data. You can also use the information to develop or identify a relationship between variables. Once the data is cleaned and prepared, you can perform different types of analysis on the data. Let us look at how you can do this.

Phase Three: Plan the Model

During this phase, you need to identify the techniques and methods to help you draw the relationship between the different variables in the data set. These relationships will help you determine the algorithms you can use in the next phase of the lifecycle. To do this, you need to apply exploratory data analytics methods and tools using various formulae and visualization methods. Let us look at some tools used for this below:

- **R** : This programming language has various modeling capabilities. It is also a good platform to use to develop the right models if you are a beginner.
- **SQL** : SQL provides a set of methods used to perform analysis within the database using different predictive models and mining functions.
- **ACCESS or SAS** : These tools can be used to access data from various storage platforms, like Hadoop, and use that data to create a reusable and repeatable model.

The market has numerous tools you can use to develop modeling techniques, but R is commonly used. At the end of this phase, you will have the required insights in your data that will help you determine the algorithm to use. The next phase is where you apply this algorithm and develop the model.

Phase Four: Build the Model

Now that you have decided which algorithm to use, you need to split the data set into training and testing data sets. In this phase, you need to consider the existing tools and determine if they are sufficient for the purpose of building a model. Make sure you identify a robust

environment to run the models. You need to analyze different techniques, such as clustering, classification, and association, to develop the model. You can use different tools to build the model.

Phase Five: Operate the Model

In this phase, you run the data through the model and deliver the reports and necessary technical documents. Additionally, you may also need to run the model in the production environment to test if it works the way it needs. This gives you an idea of how the model performs on real-time data. You can also determine any constraints in the model.

Phase Six: Communicate the Results

It is important to evaluate if the model has given you the results you needed. You can do this by analyzing your hypotheses. This is the last phase of the data science lifecycle and is where you identify the key findings and communicate the same to the organization. You can determine the results of the model based on the criteria you identified in the first phase.

III. Exercises

1. Exercise 1: Vocabulary Matching

Match the terms in Column A with their correct definitions in Column B.

Column A

- i. Lifecycle
- ii. Analytical Sandbox
- iii. ETL
- iv. Hypothesis
- v. Real-time Data

Column B

- A. A structured approach consisting of phases to complete a data science project.
- B. A secure environment for testing and analyzing data.
- C. The process of extracting, transforming, and loading data into a system.
- D. A proposed explanation or prediction that can be tested.
- E. Data that is processed and analyzed as it is generated.

2. Exercise 2: True or False

Read the statements below and mark them as **True** or **False**.

- a. The first phase of the data science lifecycle is building the model.
- b. R and Python are commonly used for data preparation and visualization.
- c. Training and testing data sets are split during the planning phase.
- d. The purpose of the discovery phase is to frame the problem and form a hypothesis.
- e. Communicating results is the last phase of the lifecycle.

3. Exercise 3: Fill in the Blanks

Complete the sentences using the correct words from the text.

- a. In the _____ phase, you identify the business requirements and frame the problem.
- b. During data preparation, you process, _____, and condition the data before modeling it.
- c. _____ is a programming language often used for modeling and analysis.
- d. The _____ phase involves running the model on real-time data and delivering reports.
- e. The results of the model are communicated during the _____ phase.

4. Exercise 4: Short Answer Questions

Answer the following questions in 1–2 sentences.

- a. What are the key goals of the discovery phase?
- b. Why is an analytical sandbox important in the data preparation phase?
- c. What tools are commonly used for planning the model?
- d. What is the purpose of splitting data into training and testing sets?
- e. How do you evaluate the success of a model in the last phase?

5. Exercise 5: Discussion

Write a short paragraph (5–6 sentences) to answer the following question:

Why is it important to communicate the results of a data science project effectively?

Include examples from the text to explain how key findings can influence business decisions.

TOPIC 3

PROS AND CONS OF DATA SCIENCE

I. Vocabulary

Word	Meaning	Word	Meaning
Ambiguous		Versatile	
Domain Knowledge		Diagnoses	
Redundancy		Tumor Detection	
Recommendation System		Career Growth	
Supervised Learning		Personal Growth	
Unsupervised Learning		Statistical Model	
Reinforcement Learning		Machine Learning Algorithm	
Genomic Sequence		Customer Behavior	
Molecular Biology		Historical Data	
Data Privacy		Data Leak	
Informed Decision		E-commerce	
Prestigious		Consultancy Services	
		Privacy Breach	

II. Reading Text

Data science is an upcoming field, and there are multiple opportunities. Having said that, this field has its share of pros and cons. This chapter looks at some pros and cons of data science, to help you take the necessary course of action.

Pros

There are many advantages to data science, and this section lists them.

Fastest Growing Field

Data science is an upcoming field and is quite in demand. If you want to become a data scientist, now is the time to do it!

Abundance of Roles

Only some people have the skills necessary to become a data scientist. You need to learn different skills and constantly learn if you want to survive in the field. This makes the field less saturated when compared to other machine learning and big data jobs. You have a lot of

opportunities if you want to switch to the field of data science. There is a very low supply of data scientists.

A Versatile Field

Data science can be applied in numerous fields, but is often used in healthcare, consultancy services, e-commerce industries and banking. Data science is versatile, and you have the chance to work in different fields.

Makes Data Easier to Use

Every company needs skilled employees to collect, process, analyze and visualize the data they collect. These employees are data scientists and they not only analyze the data but also improve the quality of that data. A data scientist knows what to do to improve and enrich the data that helps the company make informed decisions.

A Prestigious Career

A data scientist allows a company to make the right decisions. Many companies have hired data scientists to provide the right information they can use to make informed decisions. This gives a data scientist an important position in the organization. Since most companies are looking for data scientists, you can earn lot of money. According to Glassdoor, you may earn close to \$160,000 per annum.

Remove Redundancy

Data science is used in different industries, and most algorithms used in data science help to reduce the number of redundant tasks performed by people. Most companies collect historical data, and they can use this data to train machines to perform redundant tasks, which helps to simplify some jobs performed by people.

Products Become Smarter

Data science is a field that involves the use of machine learning. There are different types of algorithms used in machine learning, known as supervised, unsupervised and reinforcement learning, which look at data sets to determine customer's behavior. For instance, most e-commerce websites use recommendation systems to provide insights to the customers based on their purchase history. This has made it easier for computers to understand how humans behave.

Save Lives

The healthcare sector uses data science to improve diagnoses and predictions made about patients. Through the use of machine learning algorithms, the healthcare sector has found a way to detect tumors and cancer at an early stage. There are many other benefits the healthcare industry gains from using data science.

Aid in Personal Growth

Data science is not only a great career, but it also helps you grow in your career and personally. If you choose to become a data scientist, you will develop the right attitude and thought process to solve problems. Since the field of data science is a mixture of management and IT, you learn from both areas of business.

Cons

Data science is a hot career, and many people choose to become a data scientist because it pays well. That said, there are some disadvantages to the field. If you want to better understand data science, you also need to look at the disadvantages of data science.

Data Science is an Ambiguous Term

There is no definite definition or meaning of data science. It has become a term people use often to talk about analysis, so it is hard to know what data science actually is and what a data scientist can do. The role of a data scientist is dependent on what the company does.

Impossible to Master Data Science

As mentioned earlier, data science is a mix of numerous fields, such as computer science, mathematics and statistics. It is impossible to master the fields used in data science, which means you cannot become an expert in those fields. While most online courses have been doing their best to fill the gap that people in the data science industry are facing, it is impossible to do this. People with a background in statistics may not necessarily have the necessary information in computer science. This field changes frequently, and you need to keep learning different areas of data science if you want to stay abreast.

Requires a Lot of Domain Knowledge

From what you have read in the previous chapter, you know you need a lot of domain knowledge. If you have enough knowledge about computer science, statistics, and mathematics, you may not easily solve a data science problem if you do not have background information. The same can be said for the reverse, as well. Let us assume you work for a health-care company where you need to analyze genomic sequences. To do this, you need to have some information about molecular biology and genetics. This is the only way you can make calculated decisions that help the company. If you do not have this background, it will be difficult for you to work on analyzing genomic syndromes.

Unexpected Results

Data scientists analyze the information in the data set and use the patterns and variables in the data set to make informed decisions. This helps you make informed decisions. There are times when the data provided is arbitrary and you may not obtain expected results. The results may also vary due to the poor utilization of resources and weak management of data.

Lack of Data Privacy

Data is the new oil for numerous industries, and most companies hire data scientists to use the data they collect to make informed decisions. Having said that, the data used in these processes may lead to a privacy breach. The personal data of most clients is stored by parent companies, and some companies do not have enough security to prevent any data leaks. Many countries now have issued regulations and guidelines to prevent such leaks and secure the data.

III. Exercises

1. Exercise 1: Matching Vocabulary

Match the terms in Column A with their corresponding definitions in Column B.

Column A

- i. Supervised Learning
- ii. Redundancy
- iii. Domain Knowledge
- iv. Ambiguous
- v. Privacy Breach

Column B

- A. Knowledge specific to a particular field or industry.
- B. A machine learning method where labeled data is used for training.
- C. The unnecessary repetition of tasks.
- D. The violation of data protection leading to unauthorized access.
- E. Open to multiple interpretations or unclear.

2. Exercise 2: True or False

Mark the following statements as **True** or **False**.

- a. Data science is a versatile field applied only in healthcare.
- b. Data scientists earn an average of \$160,000 annually, according to Glassdoor.
- c. Domain knowledge is not essential to solving specific data science problems.
- d. Data science uses algorithms to analyze patterns in data and make predictions.
- e. Data privacy concerns arise due to improper security measures by companies.

3. Exercise 3: Fill in the Blanks

Complete the sentences using the correct words from the vocabulary list.

- a. Many companies use _____ systems to provide product suggestions to customers.
- b. Data science requires a lot of _____ in fields like mathematics, computer science, and the specific industry.
- c. A _____ occurs when sensitive information is accessed by unauthorized parties.
- d. In _____ learning, the system identifies patterns without labeled data.
- e. The healthcare industry uses machine learning algorithms for _____ detection.

4. Exercise 4: Short Answer Questions

Answer the following questions in 1–2 sentences.

- a. What are the benefits of data science in the healthcare sector?
- b. How does data science aid in personal and career growth?
- c. Why is data privacy a major concern in the data science industry?
- d. What makes data science a prestigious and high-paying career?
- e. Why is domain knowledge crucial in solving specific problems in data science?

5. Exercise 5: Discussion or Essay

Write a short paragraph (5–6 sentences) answering the following question:

What are the pros and cons of data science, and how can professionals overcome the challenges?

Include at least two advantages and two disadvantages, with suggestions for addressing the challenges.

TOPIC 4

TECHNICAL SKILLS REQUIRED FOR DATA SCIENCE

I. Vocabulary

Word	Meaning	Word	Meaning
Leverage		Algorithm	
Collate		Declare (variables)	
Interpret		Initialize	
Visualize		Assign	
Frameworks		Logic	
Hardcore		Futuristic Events	
Boot Camp		Predictive Modeling	
Massive Open Online Course (MOOC)		Machine Instructions	
Structured Data		Data Set	
Unstructured Data		Programming Languages	

II. Reading Text

If you want to work in the field of data science, or become a data scientist, you need to have some information about different topics mentioned in this chapter. This chapter lists some technical and non-technical skills you should develop if you want to improve in this field.

Technical Skills

As a data scientist, you need to have some technical skills that help you with statistical analysis. You need to learn how to leverage and work with different frameworks and software to mine, collect, process, collate, analyze, interpret and visualize large volumes of data. You need to develop programming skills to perform such activities. An easy way to do this is to ensure you have the necessary academic background. Most data scientists have a Ph.D. or master's degree in engineering, statistics, and computer science. This is the only way to determine if they have the foundation to help them connect with different technical points that are the foundation of the practice. Many schools offer such programs to help people pursue data science.

If you do not want to go through hardcore courses, you can look at options, such as:

- Boot camps
- Massive Open Online courses (MOOC)

These programs help you develop a basic understanding of core data science subjects. These courses also provide some information that is outside of textbook learning. You will be given real-time scenarios and asked to develop models to assess and predict futuristic events. The following are some skills you need to develop.

Understanding Data

Data science is about working with and understanding different types of data. You need to understand and love working with data. The following are some questions you can answer to help you understand whether you love data or not:

- Do you know where you need to collect data from?
- Do you understand and know what data is?
- Have you determined what information you should look at?
- How frequently do you work with data?
- Have you learned how to work with structured and unstructured data?

The most important question you need to answer is whether you love working with data or not. If yes, you need to obtain certifications, so you develop the skills to become a data scientist.

Algorithms

Algorithms are sets of instructions that you write. You can use algorithms to instruct a machine to perform specific functions and tasks. Let us try to write an algorithm using that we will instruct a computer to add two numbers.

1. Identify two variables and declare it to the machine
2. Initialize the variables
3. Ask the user to assign a value to each of these variables
4. Declare another variable to hold the sum of the first two variables
5. Calculate the sum of the first two variables and assign that value to the third variable

You can also use algorithms when you solve puzzles on paper. As a data scientist, you need to understand what algorithms are and how a machine understands them since you work with algorithms to help you analyze data. As a data scientist, you also need to learn how to design different algorithms that perform the necessary functions to help you analyze data. Let us assume you need to key in 10 numbers into the system. You may enter any ten numbers and leave it to the machine to identify the set's third largest number. To do this, you should write an algorithm to help the machine identify the number. As a data scientist, you need to write the necessary logic and develop an algorithm which helps you find the third largest number.

Programming

It is important to learn different languages, such as Java, C++, R, Python, Perl, SQL and other languages. The languages used commonly are R and Python. You can collect, clean, process, organize and analyze the information in the data set which helps you work with unstructured data.

Analytical Tools

Every data scientist needs to learn how to use different analytical tools and understand the use of Software as a Service (SaaS). This understanding will help you obtain some information from the data you have cleaned and processed using different programming languages and tools, such as Hadoop, R, SAS, Spark, Hive and Pig. Different platforms allow you to improve your skills, and you can obtain certifications that will help you move ahead in your career.

Using Unstructured Data

Data scientists need to collect, process, store, manage, clean, understand and analyze the data collected from various sources. Most of these data are not structured, and they can be collected in different forms, such as images, videos, text, emails and other forms. For instance, if you work with a credit company, you need to learn how to use historical data and patterns to assist the company in identifying the customers they can trust. You also need to learn how to collect, process and analyze the data.

III. Exercises

1. Exercise 1: Matching Vocabulary

Match the words in Column A with their definitions in Column B.

Column A

- i. Algorithm
- ii. Frameworks
- iii. Data Set
- iv. Collate
- v. Leverage

Column B

- A. A structured collection of data used for analysis.
- B. A sequence of steps or instructions for solving a problem.
- C. To organize and compile data in a systematic manner.
- D. Tools, libraries, and software used for data analysis.
- E. To use something to its advantage.

2. Exercise 2: Fill in the Blanks

Complete the sentences using the correct words from the vocabulary list.

- a. Many online platforms offer _____ to help students learn data science without attending university.
- b. Data scientists need to _____ various tools to maximize their efficiency in analyzing data.
- c. In programming, before using a variable, you must _____ and _____ it with a value.
- d. Python and R are commonly used _____ in data science.
- e. A well-designed _____ helps machines make decisions based on input data.

3. Exercise 3: True or False

Write **True** or **False** for each statement.

- a. **Java and Python** are two commonly used languages in data science.
- b. **MOOCs** provide online courses that are free or low-cost for learning data science.
- c. **An algorithm** is a random set of instructions without a structured approach.
- d. **Structured data** refers to unorganized and raw information that is difficult to process.
- e. **Collating** data means gathering and organizing information systematically.

4. Exercise 4: Short Answer Questions

Answer the following questions in 1–2 sentences.

- a. Why is it important for a data scientist to understand algorithms?
- b. What is the difference between structured and unstructured data?
- c. How can online courses help someone become a data scientist?
- d. Why are Python and R popular in data science?
- e. What are some common programming languages used in data science?

5. Exercise 5: Writing Task

Write a short paragraph (5–6 sentences) answering the following prompt:

Why are programming skills essential for a data scientist, and how do algorithms help in data analysis?

TOPIC 5

NON – TECHNICAL SKILLS REQUIRED FOR DATA SCIENCE

I. Vocabulary

Word	Meaning	Word	Meaning
Business acumen		Ethical awareness	
Communication skills		Continuous learning	
Intuition		Decision making	
Problem – solving skills		Data responsibility	
Critical thinking		Cross functional projects	
Collaboration		Data governance policies	
Team Work		Regulations	
Adaptability			

II. Reading Text

While technical skills are essential for data scientists, non-technical skills play an equally important role in their success. Strong business acumen, communication skills, intuition, problem-solving abilities, and ethical awareness all contribute to making data science more impactful. A well-rounded data scientist combines both technical expertise and soft skills to drive meaningful insights and support data-driven decision-making.

Non-Technical Skills

The following are some non-technical skills you need to develop if you want to be a data scientist. These are some personal skills a user needs to develop, and these are very different from qualifications and certifications.

Strong Business Acumen

As a data scientist, you need to have strong business acumen. You need to understand various elements important to the business model. Otherwise, you cannot use the necessary tools and skills to analyze the data set. It is only when you learn and develop these skills that you can identify the various problems and solutions to develop if you want to help the business make informed decisions so that it can sustain and grow. It becomes extremely hard for you to help an organization explore various opportunities if you do not know how to code.

Communication Skills

Data scientists can understand data better than anybody else in the organization. If you want to succeed as a data scientist, you need to learn to communicate your decisions and analysis to the stakeholders. This is the only way you can make it easier for the organization to make informed decisions. It is important to learn to communicate your findings to people in the organization, especially if they are non-technical. You need to develop the necessary communication skills to communicate your findings.

Intuition

This is an important non-technical skill you need to develop as a data scientist. It is true you may already have an insight about the data when you look at it, but you may not have the ability to perceive any hidden patterns in the data set. You need to know where to look and what to look at, so you add more value to the analysis. These are the skills that make you efficient in your work. Do not worry about cramming too much information in a short time. You learn and improve as you gain experience.

Problem-Solving Skills

Data scientists must develop strong problem-solving abilities to tackle complex data-related challenges. Often, the data provided is incomplete, noisy, or inconsistent, requiring creative approaches to extract meaningful insights. A skilled data scientist needs to break down large problems into smaller, manageable tasks and apply analytical techniques to derive solutions. Problem-solving also involves selecting the right algorithms, optimizing models, and improving efficiency in data processing.

Critical Thinking

A successful data scientist should think critically when analyzing data and drawing conclusions. Critical thinking helps in evaluating assumptions, questioning results, and ensuring data-driven decisions are reliable. It also involves assessing the validity of models, avoiding biases, and determining whether a dataset truly supports a particular hypothesis. Strong critical thinking skills allow a data scientist to distinguish between correlation and causation, ensuring their insights are meaningful and actionable.

Collaboration and Teamwork

Data science projects often require collaboration with various teams, including business analysts, engineers, and domain experts. Being able to work effectively in a team ensures that data insights align with the organization's goals and are implemented successfully. Data scientists should be open to feedback, share their expertise with colleagues, and contribute to cross-functional projects. Good teamwork ensures that insights derived from data analysis are practical and drive business impact.

Adaptability and Continuous Learning

The field of data science is constantly evolving, with new tools, technologies, and methodologies emerging frequently. A data scientist must be adaptable and willing to learn new programming languages, frameworks, and machine learning techniques. Keeping up with industry trends, attending workshops, and participating in data science communities can help professionals stay ahead in this rapidly changing field. Continuous learning is essential for staying relevant and improving one's problem-solving abilities.

Ethical Awareness and Data Responsibility

Data scientists handle large volumes of sensitive and personal data. Understanding ethical implications and data privacy concerns is crucial to ensure responsible use of information. Ethical considerations include protecting user privacy, preventing bias in machine learning models, and ensuring transparency in decision-making processes. Many organizations now follow strict data governance policies, and data scientists must be aware of regulations such as GDPR (General Data Protection Regulation) and other data protection laws.

III. Exercises

1. Exercise 1: Matching Words with Definitions

Match each vocabulary word with its correct definition.

- a. ____ The ability to perceive patterns and insights in data without explicit reasoning.
- b. ____ The ability to adjust to new tools, technologies, and methodologies.
- c. ____ Ensuring data privacy, preventing bias, and maintaining transparency.
- d. ____ Coordinating with others to improve productivity and outcomes.
- e. ____ The ability to analyze and resolve complex data-related challenges.

2. Exercise 2: Fill in the Blanks

Complete the sentences using the correct vocabulary word.

- a. A strong _____ helps data scientists understand business objectives and contribute valuable insights.
- b. Data scientists must have excellent _____ to explain their findings to non-technical stakeholders.
- c. _____ is important in data science projects because professionals from different departments work together.
- d. Continuous technological advancements require _____ to stay up to date with new methods.
- e. Companies enforce _____ to ensure data is handled ethically and securely.

3. Exercise 3: True or False

Decide whether each statement is **True (T)** or **False (F)**.

- a. **Critical thinking** helps data scientists question results and validate conclusions.

TOPIC 6

TOOLS REQUIRED FOR DATA SCIENCE

I. Vocabulary

Word	Meaning	Word	Meaning
Object-Oriented Programming		Visualization	
Data Analysis		Statistical Modeling	
Artificial Intelligence		Real-Time Data	
Machine Learning		Clusters	
Big Data		JavaScript Library	
Open-Source		APIs (Application Programming Interfaces)	
Distributed Computing		Deep Learning	
Database		Data Processing	

II. Reading text

We have looked at different aspects of data science and know why you need to use data science. Let us now look at the different tools you need to work with to become a data scientist. The following are only some tools you need to use if you want to work in data science.

R Programming Language

We have referred to R earlier in the book. This is a programming language widely used by data scientists, and many organizations use this language to perform data analysis. Organizations use R widely since it is an object-oriented programming language, and it is used for the analysis of reporting, visual graphic representations and statistics.

Python Programming Language

Python, like R, is another programming language that has features similar to other object-oriented programming languages. It is used widely in data science and to develop models. Python can be used on different operating systems, and you can develop numerous applications using this language. This programming language is used to develop machine learning models and applications that use artificial intelligence.

Hadoop

As mentioned earlier, Hadoop came into picture when it became difficult for companies to store large volumes of data. This is an open-source application which allows you to store and

process big data. You can use a distributed computing framework to analyze the data. It is best to use this application if you want to store data without having to process it. You can store as much data as needed before you decide which variable you need to use.

Structured Query Language (SQL)

SQL or Structured Query Language is used for data storage. You can create new tables and databases in SQL, and the only thing you need to do is update the records in the database. You can retrieve, modify and delete the records in SQL databases and tables. You can use this language to manage the data stored in a database.

SAS

SAS or Software as Service is an important tool to understand if you wish to work as a data scientist. This tool is designed to perform various statistical and mathematical operations. Large organizations use SAS to identify or draw hidden patterns within the data set. SAS uses SAS coding language that is often used by statisticians to develop models.

The language was developed by statisticians and professionals and is based on Reliable Commercial Software (RCS). This programming language has many statistical tools and libraries that data scientists can use to perform modeling. They can also use this language to keep the records in order. SAS is not an opensource programming language. This language is costly and can only be used by large organizations. There are some packages and libraries in SAS that cannot be accessed by organizations in the base pack, and you need to spend money to update the language.

Apache Spark

Apache Spark is a popular analytics tool used by data scientists, and it is designed for stream processing and batch processing. There are numerous APIs developed by data scientists across the globe to develop a frequent path for data to flow through for any machine learning algorithm. This application is a win when compared to Hadoop and other data processing and storing applications since it works faster. It comes with numerous APIs for machine learning and deep learning which enables data scientists to obtain hidden insights in the data set. The application can stream large volumes of data and is more efficient when compared to other big data platforms. This tool can work with real-time data, and allows you to code in Java, R and Python. You can use this application when you work with clusters of data. This tool is more efficient when compared to platforms, such as Hadoop and MapReduce.

D3.js

This is a Javascript library, and you can use this to develop visualization on the Internet. There are numerous APIs of this library, and you can use various functions to make visualizations more dynamic. You can also analyze the data set collected on the Internet. One of the advantages

of using this tool is it allows you to use animated transitions to improve data visualizations. When you use this tool, you can create dynamic documents and use options to change the data used in the visualization.

You can also use this tool along with CSS to create transitory and eminent visualizations that would show you the required graphs on the Internet. This is an important tool to use in data science, and you can work with different IoT devices. The tool allows you to work with client-side interventions for data visualization and processing.

III. Exercises

1. Exercise 1: Matching Definitions

Match the following terms with their correct definitions:

- i. **Machine Learning**
 - ii. **Big Data**
 - iii. **Database**
 - iv. **Open-Source**
 - v. **APIs**
- a) A large set of structured and unstructured data.
 - b) Software that is publicly available for modification.
 - c) A structured system for storing and managing data.
 - d) A type of artificial intelligence that helps computers learn from data.
 - e) A set of protocols for enabling communication between software applications.

2. Exercise 2: Fill in the Blanks

Complete the sentences using the correct words from the wordlist:

- a. _____ enables organizations to store and analyze massive amounts of data.
- b. In _____, programmers create reusable objects that interact with one another.
- c. Companies use _____ tools to create charts and graphs for better data understanding.
- d. A _____ is essential for storing structured data in an organized manner.
- e. Developers use _____ to allow applications to interact with each other efficiently.

3. Exercise 3: True or False

Decide whether the following statements are true or false:

- a. Python and R are both object-oriented programming languages. (**True / False**)
- b. Hadoop is a paid software used only by large organizations. (**True / False**)
- c. SQL is mainly used for managing and modifying databases. (**True / False**)
- d. Apache Spark is slower than Hadoop when processing data. (**True / False**)

- e. D3.js is a Java-based software for data storage. (**True / False**)

4. Exercise 4: Short Answer Questions

- a. What is the main purpose of Hadoop in data science?
- b. How does Apache Spark improve data processing compared to Hadoop?
- c. Why is R commonly used for statistical analysis?
- d. What is the key advantage of using D3.js for data visualization?
- e. What are some common uses of SQL in data science?

5. Exercise 5: Sentence Construction

Use each of the following words to build a sentence:

- a. **Machine Learning**
- b. **Distributed Computing**
- c. **Real-Time Data**
- d. **Open-Source**
- e. **Visualization**

TOPIC 7

ALGORITHMS AND STATISTICS

I. Vocabulary

Word	Meaning	Word	Meaning
Data Scientist		Information Gain	
Probability Theory		Predictive Analysis	
Naïve Bayes		Machine Learning	
Linear Regression		Algorithm	
Logistic Regression		Hypothesis Testing	
Neural Networks		SQL (Structured Query Language)	
K-Means Clustering		Data Stack	
Decision Tree		Parametric Learning	
Entropy		Time Series Analysis	
Bayes' Theorem		Visualization	

II. Reading text

Do you need to become a Ph.D. in mathematics or statistics to become a data scientist? No, you do not. The extent to which you need to know depends on what role you want to take up in the organization. This chapter gives you an idea of all the concepts you need to know if you want to become a data scientist. As a data scientist, you first need to understand the basic concepts of probability theory and statistics. If you want to master different statistics and mathematics algorithms, you need to practice them regularly. You need to follow a top-down approach if you want to learn how to code. You need to learn how to use a data stack, familiarize yourself with real-world projects and use libraries and documentation. When you notice your lack of theoretical background, you can see the bigger picture. It is important to learn how these algorithms work. Start off with the mathematical aspects before you move onto those areas, which are more complex. This chapter has the list of statistical and mathematical concepts you need to learn.

Naïve Bayes

A Naïve Bayes classifier is an algorithm which uses the principles of probability. One of the most important principles it uses is that the value of one feature in the data set is independent from the value of other features in the data set. According to this theorem, you can predict the probability that an event will occur based on the conditions of the occurrence of that event. It is important to learn how these classifiers work, and to do this, you need to study the basics and

principles of conditional probability and probability. One of the easiest ways to understand this theorem is to study the concepts of probability. You can then check on Bayes' theorem and how to code this theorem.

Linear Regression

Linear regression is a basic form of regression, and it allows you to determine the connection between two variables in the data set. One of the variables is termed as the predictor variable and the other is known as the response variable. These variables are often continuous in nature. A simple linear regression model uses various data point sets and then extrapolates the trend of the values and how they will progress in the future. Linear regression is a common algorithm used in parametric machine learning, and in this form of learning, you can train the machine to develop a mathematical function using existing data sets. You can use this function to make predictions or forecasts about future events. These functions are termed as models in machine learning terminology. The best way to learn more about regression is to develop an understanding of elementary statistics. If you want to learn more about regression and the various concepts, you need to take an advanced course of statistics.

Logistic Regression

Logistic regression is a process used in different cases where you have a binary dependent variable. This form of regression focuses on the estimation of probability of the occurrence of an event. Logistic regression is also an algorithm used in parametric machine learning. Like linear regression, this algorithm also results in a mathematical function. The difference between these forms of regression is that logistic regression leads to developing a mathematical function that estimates the values based on their logarithms or other statistical functions. Another difference between linear and logistic regression is that the former uses real numbers and gives those as the output. The latter uses a model which provides the logistic function. A logistics function produced is also termed as a sigmoid function that takes care of the values to give you an output between 0 and 1. Let us understand why a sigmoid function returns a probability value. It is because of the algebraic concepts used in the algorithms that a negative exponent is taken.

Neural Networks

A neural network is a form of machine learning algorithm. These forms of networks are based on the structure of the neurons in the brain. The neural network uses a series of activation points or weights to help you make the necessary predictions. The neurons take the input, apply the transformation function and give you an output that helps you make the necessary decisions.

When it comes to capturing any nonlinear relationships within the data set, it is best to use a neural network best since it helps in tasks, such as image and audio processing. The fundamental concept of neural network is to transform the input and processing it to generate the output. If you

want to understand the math better in a neural network, you need to take up courses to help you learn more about linear algebra and geometry. This is a better way to start preparing. If you want to delve deeper, you need to learn more about graph theory, matrix theory and real analysis.

K-Means Clustering

The k-means clustering algorithm is a type of unsupervised machine learning algorithm. You can use this algorithm to categorize any unlabeled data. This type of data does not have any defined categories or groups, but it works to identify or define any hidden categories in the data set. The number of variables identified by the algorithm are 'k' in number. The algorithm iterates over the various data points in the data set, so every point is assigned to one of many groups. This algorithm depends on the idea that the distance between the data points and the center of the cluster is the same across all data points. You can use any function that shows the distance between the elements in the data set.

Suppose you want to work with this algorithm. In that case, you need to understand the basics of algebra, including addition and subtraction to identify the distance between the data points easily. If you want to delve deeper, you need to learn about Euclidean and non-Euclidean geometry.

Decision Tree

Decision trees are an easy way to determine the outcome of any decision you take using a flowchart. The flowchart is in the form of a tree structure that uses a branching method. Every node in the decision tree is based on a specific variable and every branch is an image of the outcome. A decision tree is based on information theory, since this is how they are constructed. An important concept of a decision tree is known as entropy. Entropy is a measure of uncertainty in any variable. When you use entropy, you can construct a decision tree easily.

When there is higher entropy, there is more uncertainty in the data. This means you need to split the tree in a way to decrease any uncertainty. Information gain helps you determine how much information you can gain from someone. When it comes to a decision tree, you can calculate the information gain on every column based on previous columns based on the information present in the columns in the data set. Here is a small tip - you need to learn basic algebra and probability to know what decision trees are and how they work. If you want to learn more about decision trees, you need to learn about logarithms and probability.

Some Final Thoughts

Mathematics and statistics are difficult subjects, and they can feel daunting and dry at times. You will, however, feel equipped when you compare your skills against the skills of your peers. It is also important to learn how you can apply these concepts on various data science problems. The topics highlighted in this book will need to be read well and understood. This is the

only way you can develop and build algorithms. You can rely on different machine learning libraries and tools to complete this work for you. It is useful for a data scientist to understand statistics and math, which helps you determine what is happening in the tools or algorithms. This allows you to choose the algorithm that works best for your data set and allow you to make the necessary predictions. Dive deep and work hard to develop the skills necessary to become a data scientist.

Applying Math to Data Science Models

If you want to work as a data scientist, you need to have some mathematical skills. This is the only way you can understand data and the significance of the patterns in the data set. These are important data science skills since you use them to build and develop models and perform predictive analysis and hypothesis testing. Data scientists use math to develop decision models and they can use different mathematical algorithms to predict the future. This book explains some concepts you need to learn and the skills you need to develop if you want to become a data scientist.

Deriving Insights from Statistical Methods

If you are a data scientist, you need to understand the basics of statistics. You should understand the significance of data, develop hypotheses and validate them by simulating certain scenarios. This makes it easier for you to predict futuristic events. It is very difficult to develop advanced statistical skills, but if you are keen about pursuing a career in data science, you need to understand some basics of logistic regression, linear regression, time series analysis, Bayes classification, etc.

Some Essentials for Data Science

Understanding programming languages and coding in various languages, especially R and Python, are important for data science. You need to learn how to write code to instruct the machine on how it should process, manipulate, analyze and visualize the data. It is important to understand how to write code in R and Python. There are different libraries and functions in these languages that allow you to manipulate, analyze and visualize the analysis of the data set.

You can also use SQL to run queries on the data set that allows you to extract and modify data in the database. People also use JavaScript library to develop interactive, dynamic and custom visualizations on the Internet. To do this, you need to learn to write code in R or python. It is important to learn to code if you want to become a data scientist, and these are easy languages to learn.

III. Exercises

1. Exercise 1: Matching Definitions

Match the terms with their correct definitions:

- i. **Neural Networks**

- c. How does a neural network process information?
- d. What is the purpose of a decision tree in machine learning?
- e. Why is visualization important in data science?

5. Exercise 5: Sentence Construction

Use each of the following words in a sentence:

- a. Probability Theory
- b. Hypothesis Testing
- c. Data Stack
- d. Logistic Regression
- e. Time Series Analysis

6. Exercise 6: Multiple-Choice Questions

1. What is the primary use of Bayes' theorem in data science?
 - a) Sorting data into categories
 - b) Predicting the probability of an event occurring
 - c) Drawing graphs for data visualization
 - d) Storing large amounts of data
2. Which of the following is NOT a machine learning algorithm?
 - a) K-Means Clustering
 - b) Linear Regression
 - c) SQL
 - d) Neural Networks
3. What is a key characteristic of a sigmoid function in logistic regression?
 - a) It predicts continuous values
 - b) It outputs a probability between 0 and 1
 - c) It classifies data into multiple categories
 - d) It does not use mathematical functions

7. Exercise 7: Identify the Algorithm

Read the descriptions below and identify which algorithm is being described:

- This algorithm assigns data points to clusters based on their distance from a central point. (_____)
- This classification algorithm is based on conditional probability. (_____)
- This regression model is used to estimate the likelihood of an event occurring. (_____)
- A supervised learning algorithm that uses a flowchart-like structure to make decisions. (_____)
- A mathematical model that predicts future values based on past data trends. (_____)

8. Exercise 8: Coding Challenge (Conceptual)***

Describe, in simple terms, how you would write a Python program to:

- a. Implement linear regression using a dataset.
- b. Classify spam emails using a Naïve Bayes classifier.
- c. Predict future stock prices using time series analysis.

9. Exercise 9: Concept Application ***

Explain how each of the following concepts can be applied in real-world data science projects:

- a. Decision trees in medical diagnosis

- b. K-Means clustering in customer segmentation
- c. Logistic regression in fraud detection
- d. Neural networks in image recognition
- e. SQL in managing e-commerce databases

10. Exercise 10: Discussion Question***

Mathematics and statistics play a crucial role in data science. Based on the topics covered, discuss the importance of these fields in developing machine learning models and making data-driven decisions.

TOPIC 8

DATA SCIENCE IN A SECTOR

I. Vocabulary

Word	Meaning	Word	Meaning
Statisticians		Cloud-based solutions	
Data Science		Big Data	
Logistic regression		Open source	
Healthcare industry		Experts	
Insights		Consumption	

II. Reading text

Most statisticians have been stating that data science is exactly what they have been doing for the last few decades. While this may be true to an extent, you cannot agree with them. Data science is completely different and separate from traditional statistical and mathematical approaches. Data science is different from statistics in many ways. Data scientists use programming languages to improve the efficiency of their analysis. If you are a statistician, you need to know the subject in its entirety. As a data scientist, you need only to understand the basics of mathematics and statistics.

Statisticians have very limited expertise about anything apart from statistics. They need to reach out to subject matter experts if they want any help or information about other subjects. They also need some help from experts to help them determine and understand their findings. This is the only way they can learn how they can move forward.

On the other hand, a data scientist has enough information about different subjects that helps him understand and analyze the findings with ease. A data scientist generates deep insights and uses his subject matter expertise to understand what the data is trying to tell him. The following are some ways to use data science to improve the performance of the business in specific sectors.

- Some architectural firms have begun to use machine learning algorithms to optimize the consumption of energy in modern designs.
- Marketing firms and businesses use logistic regression and other statistical algorithms to predict how customers may behave in the future.
- The healthcare industry uses data science to determine illnesses and personalize treatment plants. This allows doctors to predict and check for any future health problems.
- A data journalist takes information from the website to identify any patterns and connections within the data set.

- Law enforcement agencies and organizations use data science to prevent and predict any criminal activities

Communicate the Insights Captured Through Data

If you want to become a good data scientist, you need to have sharp written and oral communication skills. As a data scientist, you need to communicate the information you have inferred from the data. If you cannot do this, all the information in the data set means nothing to you or the business. A data scientist must find a way to explain the insights gathered from the data set so that he can develop meaningful and clear visualizations. It is important to remember that people are visual, so they learn better when they can visualize anything. As a data scientist, you need to be pragmatic and creative about the data collected and find the right way to communicate them clearly.

Leverage Cloud-Based Solutions

If you are not used to working with large volumes of data, you can use cloud-based solutions to learn more about the patterns in the data set. Most organizations and data analysts have understood the use of data analytics and big data and their importance. It is a new domain for some people and organizations, but they are working as hard as possible to prepare for the flood of big data. many cloud applications, both private and new, are dedicated to improving the way companies handle their big data. You can also use cloud services, such as Tableau, to clean and process data sets.

Most cloud-based applications are open-source and code-free. You can use these solutions to model and visualize data using various statistical tools and techniques. You need basic knowledge of mathematics and statistics before you apply any data science techniques on the data set. If you do not have knowledge about programming, you can use applications that do not require you to write any scripts or code.

Regardless of whether you use cloud-based solutions to perform data analysis or not, you need to train your employees and gather more information about what data science is. You also need to learn about the algorithms discussed earlier. It is important for you to learn more about how to process data, design a model, build the model and analyze it if you want to obtain insights about the data. You cannot expect any cloud-based platform to make it easier for anybody to work with the data set if they do not have basic knowledge of mathematics and statistics.

III. Exercises

1. Exercise 1: True or False

Determine whether the following statements are **true** or **false**:

- Data science and statistics are exactly the same.
- A data scientist needs deep expertise in only one subject.

- c. Machine learning algorithms can help optimize energy consumption in architecture.
- d. Data scientists must communicate their insights clearly.
- e. All cloud-based applications require programming skills.

2. Exercise 2: Fill in the Blanks

Complete the sentences using words from the **word list**:

- a. _____ is used to predict customer behavior using statistical algorithms.
- b. Doctors use _____ to personalize treatment plans.
- c. A _____ scientist must know how to present data insights effectively.
- d. Many _____ applications help organizations analyze large datasets.
- e. _____ is a way to represent data graphically.

3. Exercise 3: Short Answer

Answer the following questions in **one or two sentences**:

- a. How do machine learning algorithms help businesses?
- b. Why is visualization important for data scientists?
- c. What role does cloud computing play in data science?
- d. What are some industries that benefit from data science?
- e. How is data science different from traditional statistics?

4. Exercise 4: Matching

Match each **term** with its **correct definition**:

Term	Definition
i. 1. Big Data	A. Used to predict customer behavior.
ii. 2. Logistic Regression	B. Helps process and analyze large datasets.

Term	Definition
iii. 3. Machine Learning	C. A statistical model for binary outcomes.
iv. 4. Cloud-Based Solutions	D. Represents data graphically.
v. 5. Visualization	E. Algorithms that enable computers to learn from data.

5. Exercise 5: Writing Exercise

Write a short paragraph (5–6 sentences) answering one of the following questions:

- a. Explain how data science helps improve the healthcare industry.
- b. Compare and contrast the roles of a statistician and a data scientist.
- c. Why is cloud computing important for businesses using big data?

THE END!!!